

Module 02 Lab Exercise - Reflection Journal

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ITAI 1371 - Introduction to Machine Learning

Completing this lab helped me gain a better understanding of how data is organized, analyzed, and visualized when working with machine learning. Before beginning this assignment, I already had some experience using Python, NumPy, Pandas, and Matplotlib from previous assignments. I was familiar with concepts such as calculating the mean and median and creating basic graphs. However, one area that I initially struggled with during this lab was understanding the dataset itself. Seeing terms such as samples, features, classes, and dataset shape together was somewhat overwhelming at first. As I worked through each section of the lab, I began to understand what those terms represented and how they related to the information being analyzed.

Working with the Iris dataset helped make these concepts clearer to me. I learned that the dataset contained 150 flower samples, four different features, and three flower species. Understanding that the rows represented individual samples while the columns represented different measurements helped me see how a dataset is structured. Converting the information into a Pandas DataFrame also made the data easier for me to understand because I could see the information organized into a table rather than looking at the data as just numbers in Python.

One of the most useful parts of this lab for me was creating and interpreting the scatter plot. Instead of only looking at rows of measurements, I could visually see patterns in the data. The Setosa flowers appeared more separated from the other species, while Versicolor and Virginica had more overlap when comparing sepal length and sepal width. This helped me understand why data visualization is important. A graph can make relationships and patterns easier to recognize than simply reading raw values. It also made me wonder whether using petal length and petal width would make the differences between the species even easier to identify.

The statistical portions of the lab also helped strengthen concepts that I have previously learned. I used NumPy to calculate the mean and standard deviation of the sepal lengths and used Pandas to group the flowers by species and compare their average measurements. I was already familiar with calculating averages, but this assignment helped me understand how those calculations can be applied to an actual dataset. I also learned that standard deviation provides information about how much the values vary around the average. Seeing these concepts applied to real data made them more meaningful than simply calculating numbers by themselves.

Overall, this lab helped me become more comfortable working with datasets and understanding the purpose behind the different steps of data analysis. My biggest challenge was initially understanding what the dataset was showing me, but working through the lab step by step helped the information become much clearer. I now have a better understanding of how NumPy, Pandas, and Matplotlib can work together to analyze data, calculate statistics, and create visualizations. I also have a better understanding of how these skills connect to machine learning because the computer needs

organized and understandable data before it can recognize patterns or make predictions. I still have more to learn, but I feel more confident reading and interpreting a dataset than I did before completing this assignment.